FIITJEE

ALL INDIA INTEGRATED TEST SERIES

HALF COURSE TEST – V

JEE (Main)-2021

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 75 questions.
- Each subject (PCM) has 25 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only three sections: **Section-A**, **Section-B and Section-C**.

  Section-A (01 – 20, 26 – 45, 51 – 70) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 22, 46 – 47, 71 – 72) *contains 6 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

  Section-C (23 – 25, 48 – 50, 73 – 75) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

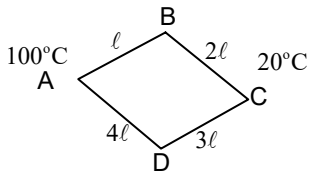
PART – I

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. There are two coherent sources in same phase of sinusoidal spherical waves that produce an interference pattern in space. The wavelength of the wave is λ, the phase speed is v. The interference pattern is observed in the perpendicular bisector plane of line joining source. What curve do the positions of central maxima of interference patter belong to?
(A) Straight line (B) Ellipse
(C) Hyperbola (D) Circle

2. Four roads AB, BC, CD, DA of equal cross sectional area and made of different materials having thermal conductivities in ratio 1 : 2 : 3 : 4, respectively, and lengths as show. They are arranged as show in figure. Point A & C are maintained at temperatures $100^{o}C \& 20^{o}C$, respectively. Then,



(A) Rate of heat flow through AB is equal to that through DC.
(B) Temperature of D is $70^{o}C$
(C) Temperature of D is $50^{o}C$
(D) If we connect a fifth rod through BD, then temperature of B & D will charge.

3. A rescue ship tries to raise a sunken miniature submarine from the bottom of a lake. The submarine and its contents have a mass of 72000 kg & volume $18.9m^3$. What upwards force must be applied to raise the submarine? (The density of water is 1000 kg/m^3. Take $g = 9.8m/s^2$):
(A) $1.8\times10^5 N$ (B) $2.0\times10^5 N$
(C) $4.8\times10^5 N$ (D) None of these

Space for Rough work

4. A steady non-turbulent stream of water comes out of a tap and falls vertically downward. As it does so, the diameter of the stream appears to get smaller. What is the primary reason?
(A) The water's surface tension constricts the stream
(B) Air pressure, which decreases with altitude, squeezes the stream
(C) The water is accelerating under gravity and so the stream must get thinner as the volume flux (velocity times cross-sectional arc) must remain constant
(D) The flow does not constrict, it is on optical illusion

5. 56 turning forks are so arranged in series that each fork gives 4 beats per sec with the previous one. The frequency of the last fork in 3 times that of the first. The frequency of the first fork is:
(A) 110 (B) 56
(C) 60 (D) 52

6. Consider two uniform spherical planets of equal density but unequal radius. Which of the following quantities is the same for both planets?
(A) The escape velocity from the planet's surface
(B) The acceleration due to gravity at the planet's surface
(C) The orbital period of a satellite in a circular orbit just above the planet's surface
(D) The orbital period of a satellite at a given distance from the planet's centre

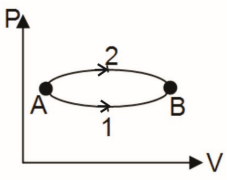
7. Two streams of water flow through the U-shaped tubes shown. The tube on the left has cross – sectional Area A, and the speed of the water flowing through it v; the tube on the right has cross – sectional area $A' = \frac{1}{2}A$. If the net force on tube assembly is zero, what must be the speed v' of the water following through the tube on the right? Neglect gravity, and assume that the speed of the water is each tube in the same upon entry and exit:



(A) $\frac{v}{2}$ (B) v
(C) $v\sqrt{2}$ (D) $2v$

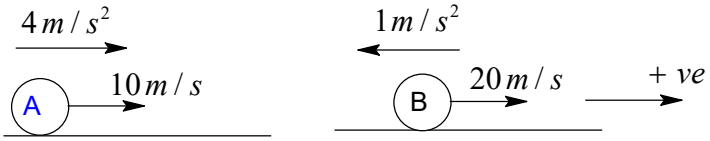
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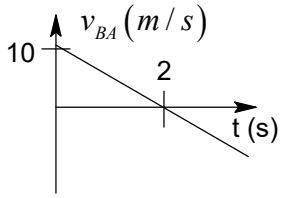
8. The figure shows two paths for the change of state of a gas from A to B. The ratio of molar heat capacities of path 1 and path 2 is

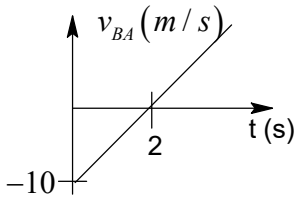


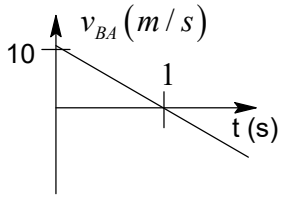
(A) > 1 (B) < 1
(C) 1 (D) data insufficient

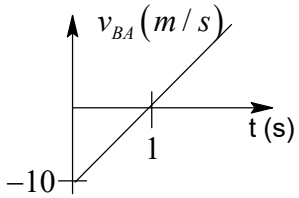
9. Initial velocity and acceleration of two particles are as shown in figure. Assuming the shown direction as the positive v_{BA} *versus* time graph is as:



(A)


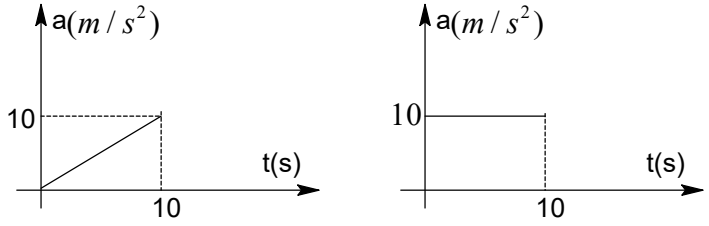
(B)


(C)


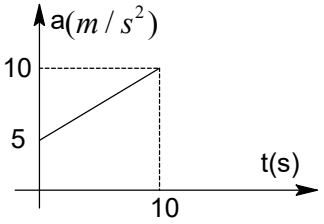
(D)


Space for Rough work

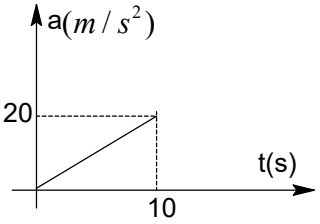
10. Acceleration of two identical particles moving in a straight line is as shown in figure. The corresponding $a-t$ graph of their centre of mass will be:



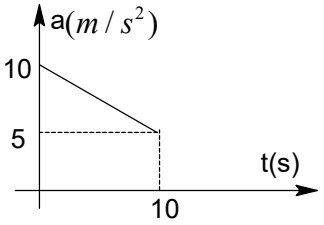
(A)



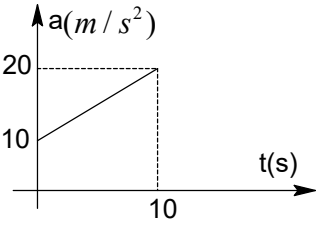
(B)



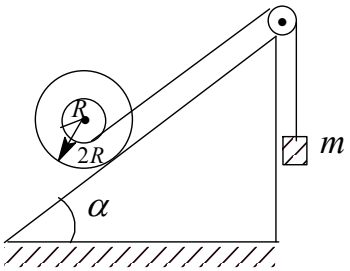
(C)



(D)



11. A spool of mass M and radius 2R lies on an inclined plane as shown in figure. A light thread is wound around the connecting tube of the spool and its free end carries a weight of mass m. The value of m so that system is in equilibrium is: (friction in sufficient)

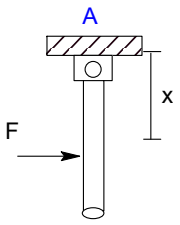


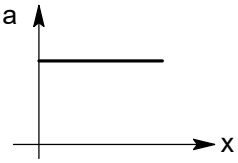
(A) $2M\sin\alpha$ (B) $M\sin\alpha$

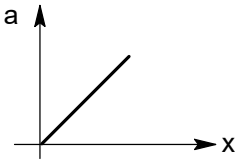
(C) $2M\tan\alpha$ (D) $M\cos\alpha$

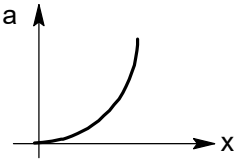
Space for Rough work

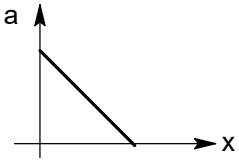
12. A rod of mass m and length l is hinged at one of its end A as shown in figure. A force F is applied at a distance x from A. The acceleration of centre of mass (A) varies with x as:



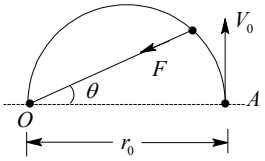
(A) 

(B) 

(C) 

(D) 

13. A particle of mass m is projected from point A with an initial velocity V_0 perpendicular to line OA and moves under a central force F along a semicircular path of diameter $OA = r_0$ its velocity at an angle θ: (Assume no other force then S)



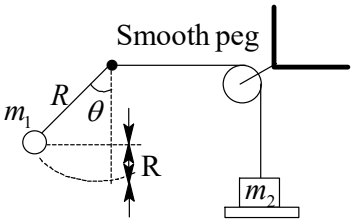
(A) $V_0\cos\theta$

(B) $\dfrac{V_0}{\cos\theta}$

(C) $\dfrac{V_0}{\cos^2\theta}$

(D) $\dfrac{V_0\sin 2\theta}{2}$

14. Two blocks are connected by a ideal string that passes over a frictionless peg as shown in figure. One end of the string is attached to a mass $m_1 = 3$ kg, i.e., a distance $R = 1.20m$ from the peg. The other end of the string is connected to a block of mass $m_2 = 6kg$ resting on a table. From what angle θ, measured from the vertical, must the 3 kg block be released in order to just lift the 6 kg block off the table:



(A) $60°$

(B) $90°$

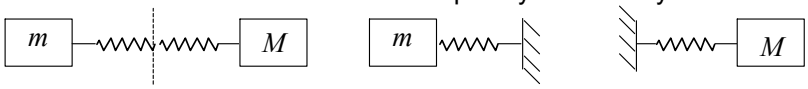
(C) $70°$

(D) $75°$

Space for Rough work

15. A uniform spherical shell is gradually expanding maintaining its shape. The gravitational potential at its centre:
(A) Increases (B) Decreases
(C) Cannot be determined (D) Remains same

16. 2 blocks of mass m and M are connected by a spring of spring constant K and the system is vibrating with certain frequency. What position from the block of mass m should the spring cut so that the blocks oscillate with same frequency individually?



$(\text{Length of spring}=L)$

(A) $\frac{L}{2}$ (B) $\frac{mL}{m+M}$
(C) $\frac{ML}{m+M}$ (D) not possible

17. An object of mass *m* is moving in a uniform circular motion in the *xy* plane. The circle has radius *R* and the object is moving around the circle with speed *v*. The motion is projected onto the x axis where it appears as simple harmonic motion according to $x(t) = R\cos(\omega t + \phi)$, here ω is equal to
(A) v/R (B) m^2R
(C) R/v (D) $v/(R\sin \omega t)$

18. Water at 50°C is filled in a closed cylindrical vessel of height 10 cm and cross sectional area 10 cm^2. The walls are adiabatic but flat parts are made of 1 mm thick aluminium $(K=200\,J/ms°C)$. Assume that outside temperature is 20°C. The density of water is 10^3 kg/m^3 and specific heat of water is $4200\,J/kg°C$. Time taken for temperature to fall by 1°C (Assume that heat is lost due to conduction only):
(A) 1 sec (B) 10 sec
(C) 35 sec (D) 70 ms

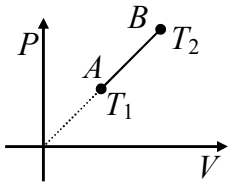
Space for Rough work

19. PV diagram of a diatomic gas is a straight line passing through origin. Molar heat capacity of gas in the process will be:

(A) $4\mathrm{R}$ (B) $2.5\mathrm{R}$

(C) $3\mathrm{R}$ (D) $4\mathrm{R}$

20. Two moles of an ideal monatomic gas are taken from state *A* to state *B* through a process as shown in figure. The work done by the gas in the process is



(A) $R(T_2 - T_1)$ (B) $2R(T_2 - T_1)$

(C) $\frac{1}{2}R(T_2 - T_1)$ (D) $\frac{1}{4R}(T_2 - T_1)$

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. A pendulum is suspended from the ceiling of a car which is moving up over an inclined plane of inclination 30º with horizontal with some upward acceleration of. Equilibrium position of the pendulum is at an angle of $\tan^{-1}\frac{\sqrt{3}}{5}$ from the vertical, what is the acceleration of the car?

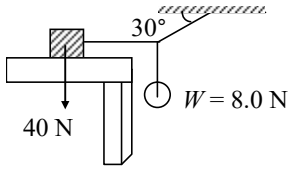
22. Two particles are performing simple harmonic motion with same amplitude and same frequency. When they are at same distance from mean position on opposite sides, their speeds are one fourth of maximum speed. Also at these positions their direction of velocities are opposite. If the phase difference between the two simple harmonic motions is $\frac{3\pi}{x}$, then find the value of x.

Space for Rough work

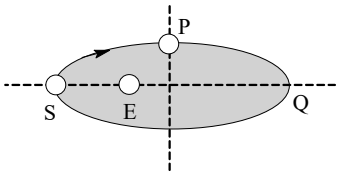
SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

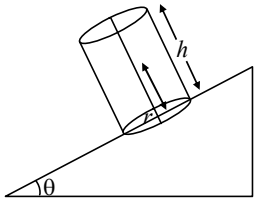
23. The system shown in figure is just on the verge of slipping. The coefficient of static friction between the block and table is



24. The satellite is moving in an elliptical orbit about the earth as shown in the figure. The minimum and maximum distances of satellite from earth are 3 units and 5 units, respectively. The distance of satellite from the earth when it is at P is equal to (in units)



25. A cylinder of height 20cm rests on an inclined plane. As the angle of inclination increases, the cylinder toppled at an angle of 45º. The radius of the cylinder will be (in cm)



Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

26. Which of the following statements is incorrect about significant figures?
(A) All non-zero digits are significant
(B) Zeroes preceding to first non-zero digits are not significant.
(C) Zeroes between two non-zero digits are not significant.
(D) Zeroes at the end or right of a number are significant provided they are on the right side of the decimal point.

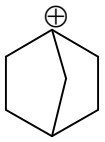
27. The compound that contains the most acidic hydrogen is:
(A) $CH_2 = CH_2$ (B) $HC \equiv CH$
(C) $CH_2 = C = CH_2$ (D) $CH_3 - CH_3$

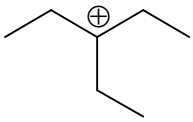
28. With respect to periodic properties, the correct statement is:
(A) Electron affinity order is $F > O > Cl$
(B) First ionisation energy order is $Al > Mg > K$
(C) Atomic radius order is $N > P > As$
(D) Ionic radius order is $K^+ > Ca^{2+} > Mg^{2+}$

29. Consider the following four xenon compounds:
XeF_2, XeF_4, XeF_6 and XeO_3. The pair of xenon compounds expected to have non-zero dipole moment is:
(A) XeF_4 and XeF_6 (B) XeF_2 and XeF_4
(C) XeF_2 and XeO_3 (D) XeF_6 and XeO_3

Space for Rough work

30. The correct order of stability of the following carbonium ions is:


[I]

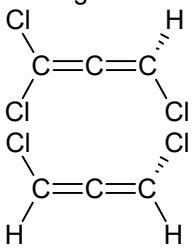

[II]

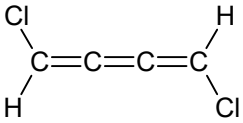

[III]

(A) II>I>III
(B) III>II>I
(C) I>III>II
(D) II>III>I

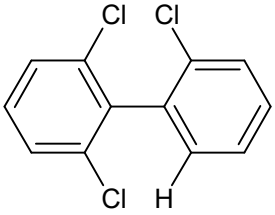
31. Among the following, the element having maximum inert pair effect is:
[Given: Atomic number of Ge = 30, Pb = 82, Si = 14 and Sn = 50]
(A) *Ge*
(B) *Pb*
(C) *Si*
(D) *Sn*

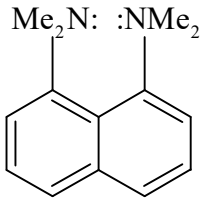
32. Which among the following molecules is chiral?

(A) 

(B) 

(C)

(D) 

33. 

$1,8-\text{Bis}(\text{dimethylamin})\text{napthalene}$

The basic strength of above compound is 10^{10} more than 1 – dimethyl amino naphthalene. Reason for high basic strength is:
(A) Lone-pair lone pair repulsion is minimized in conjugate acid of the base.
(B) Inductive effect
(C) Due to presence of two lone pairs there is more electron density, is the reason for its higher basic strength
(D) Hyperconjugation

Space for Rough work

34. The diatomic molecule that has two π – type bonds is:
(A) B_2 (B) F_2
(C) N_2 (D) O_2

35. A 10L flask containing 10.8 g of N_2O_5 is heated to 373K, which leads to its decomposition according to the equation $2N_2O_5(g) \rightleftharpoons 4NO_2(g) + O_2(g)$. If the final pressure in the flask is 0.5 atm, then the partial pressure of $O_2(g)$ in atm is:
(A) 0.06 (B) 0.04
(C) 1.2 (D) 0.24

36. Acetic acid content is commercial vinegar was analysed by titrating against 1.5M NaOH solution. A 20 mL vinegar sample required 18 mL of titrant to give end point. The concentration of acetic acid in the vinegar (in mol L^{-1}) is:
(A) $1.35\,mol/L$ (B) $1.45\,mol/L$
(C) $2.45\,mol/L$ (D) $3.45\,mol/L$

37. At 25°C dissociation constants of acid HA and base BOH in aqueous solution are same. The pH of 0.01M solution of HA is 5. The pOH of $10^{-4}M$ solution of BOH at the same temperature is:
(A) 3.5 (B) 4
(C) 6 (D) 9

38. The average value of C – C bond order in graphite is:
(A) 1 (B) 3/2
(C) 3/4 (D) 4/3

39. The temperature (T) dependence of the equilibrium constant (K) of a chemical reaction is correctly described by the following statement:
(A) For an endothermic reaction, the slope of lnK v/s 1/T plot is positive.
(B) For an exothermic reaction, K is directly proportional to T.
(C) For an exothermic reaction, K at a higher temperature is lower than K at a lower temperature.
(D) If ΔH is independent of temperature the change in K is also independent of temperature.

Space for Rough work

40. The molar heat capacity at constant volume of a colourless gas is found to be 25J mol^{-1} K^{-1} at room temperature. The gas must be:

(A) N_2 (B) O_2

(C) CO_2 (D) SO_2

41. The compressibility factor $Z=\left(\frac{PV}{nRT}\right)$ of a gas above $T=\left(\frac{8a}{27Rb}\right)$ will be:

(A) less than unity (B) greater than unity

(C) equal to unity (D) equal to 2

42. The rate of nitration of the following aromatic compounds decreases in the order:

(i) benzene (ii) pyridine

(iii) Thiophene (iv) toluene

(A) $iv > i > iii > ii$ (B) $iii > iv > i > ii$

(C) $iii > ii > i > iv$ (D) $ii > i > iv > iii$

43. Which of the following statement is wrong?

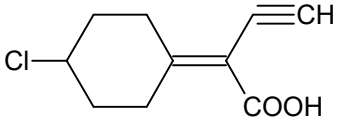
(A) Polar stratospheric clouds $(PSCs)$ are clouds formed over Antarctica

(B) Acid rain dissolves heavy metals such as Cu, Pb, Hg and Al from soil, rocks & sediments.

(C) H_2SO_4 is major contributor to acid rain, HNO_3 ranks second.

(D) Fishes grow as well in warm as in cold water.

44. What are the correct designations for the structure below?



(A) E, E (B) Z, E

(C) E, Z (D) No geometrical isomers are possible

45. Each of the following statements is correct except:

(A) The lattice positions left vacant by anion and occupied by electrons are called F-centres

(B) The presence of F-centres makes the lattice electrically neutral and diamagnetic

(C) The presence of F-centres gives colour to the crystals

(D) F-centres contain unpaired electrons.

Space for Rough work

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

46. The total number of bonds in the elemental white phosphorus molecule is:

47. Specific rotation of the (R)-enantiomer of a chiral compound is $48°$. The specific rotation of a sample of this compound which contains 25% of (S)-enantiomer is $(6x)°$. Then the value of x is

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

48. The effective nuclear charge of He atom is 1.7. The first ionisation energy of He atom in eV is:

49. The following reaction is carried out at 1 atm & 300K.

$$2H_2(g) + O_2(g) \longrightarrow 2H_2O(\ell)$$

ΔU for the above reaction is 550 KJ. Assuming ideal gas behaviour for H_2 and O_2. Calculate the value of ΔH (KJ). The value of gas constant, R = 0.082 L atm K^{-1} mol^{-1} = 8.314 $JK^{-1}mol^{-1}$: [Given: The volume of 1 mol of liquid water is 18mL under the above reaction condition]

50. A mixture of C_3H_8 and oxygen in 1L closed vessel has an internal pressure of 4 atm at 100°C. When the mixture is ignited, the reaction produces CO_2(g) and H_2O(g) until all the oxygen is consumed. After the reaction, pressure of the vessel is 4.2 atm at the same temperature. Calculate the weight (in gm) of oxygen present before the reaction. [Gas constant, R = 0.082 L atm mol^{-1} K^{-1}]

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

51. If $(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_{nx^n}$. Then $C_1 + 2C_2 + 3C_3 + ... + nC_n$ is

(A) 2^{n-1} (B) 2^n

(C) $n.2^{n-2}$ (D) $n.2^{n-1}$

52. The complex number z, which lies on the common tangent to the curves $|z-2i|=2$ and $|z-1-3i|+|z+1-3i|=2\sqrt{3}$ will satisfy

(A) $\mathrm{Re}(z)=4$ (B) $|z|=4$

(C) $\mathrm{Im}(z)=4$ (D) $\mathrm{Re}(z)+\mathrm{Im}(z)=4$

53. The value of ${}^{404}C_4 - {}^4C_1\,{}^{303}C_4 + {}^4C_2.{}^{202}C_4 - {}^4C_3\,{}^{101}C_4$ is equal to

(A) $(401)^4$ (B) $(101)^4$

(C) 0 (D) $(201)^4$

54. Sum of the infinite series $\frac{4}{1!}+\frac{11}{2!}+\frac{22}{3!}+\frac{37}{4!}+\frac{56}{5!}+.....\infty$ is

(A) $6e+1$ (B) $3e+2$

(C) $e-6$ (D) $6e-1$

Space for Rough work

55. If $x^n + x - 1 = 0$ is divisible by $x^2 - x + 1 = 0$ for some n > 2, then the remainder when 6 divides n is
(A) 0 (B) 2
(C) 3 (D) 5

56. Number of non – negative integral solutions (x, y) of the equation $y^2 + 6xy - 8x = 0$ is
(A) 0 (B) 1
(C) 3 (D) Infinite

57. A square $OABC$ is formed by line pair $xy = 0$ and $xy + 1 = x + y$ where 'O' is the origin. A circle with centre C_1 inside the square is drawn to touch the line pair $xy = 0$ and another circle with centre C_2 and radius twice that of C_1, is drawn to touch the circle C_1 and the other line pair. The radius of the circle with centre C_1 is:
(A) $\frac{\sqrt{2}}{\sqrt{3}\left(\sqrt{2}+1\right)}$ (B) $\frac{2\sqrt{2}}{\sqrt{3}\left(\sqrt{2}+1\right)}$
(C) $\frac{\sqrt{2}}{3\left(\sqrt{2}+1\right)}$ (D) $\frac{\sqrt{2}+1}{3\sqrt{2}}$

58. If $x^2 - 2axy - y^2 = 0$ and $x^2 - 2bxy - y^2 = 0$ bisect the angles between each other, then the relation between 'a' and 'b' is:
(A) $a + b = 1$ (B) $ab = 1$
(C) $ab + 1 = 0$ (D) $a^2 + ab + b^2 = 0$

59. Let $x + \alpha y - 1 = 0, \alpha \in R$ be a variable focal chord of parabola $y^2 = 4x$, which cuts the parabola at A and B. If normal at A and B meet at C, then locus of C may be
(A) $x + y^2 + 3 = 0$ (B) $x - y^2 + 3 = 0$
(C) $x - y^2 - 3 = 0$ (D) $x + y^2 - 3 = 0$

Space for Rough work

60. A, B, C, D are consecutive vertices of a rectangle, whose area is 6 and length of diagonal is $\sqrt{13}$. Area of ellipse passing through A and C and having foci at B and D is

(A) $\frac{5\sqrt{3}\pi}{2}$ (B) $\frac{\pi}{2}$

(C) $\frac{\pi}{2\sqrt{3}}$ (D) $\frac{\pi}{\sqrt{3}}$

61. If two tangents drawn from (k, 2) to the hyperbola $\frac{x^2}{4} - y^2 = 1$ touches it in opposite quadrants then range of values of k is

(A) $(1,5) \cup (-5,-1)$ (B) $(3,4) \cup (-4,-3)$

(C) $(2,3) \cup (-3,-2)$ (D) $(2,4) \cup (-4,-2)$

62. t_1, t_2, t_3 are 3 points on $y^2 = 4x$ such that normal at t_1 intersects parabola at t_2 and normal at t_2 intersects parabola at t_3. If $3t_1 + 13t_2 + 9t_3 = 0$. Then $\frac{t_1^2 + 5}{t_2^2 - 6} =$ _____

(A) 1 (B) 3

(C) –2 (D) 5

63. The number of ways in which 21 identical balls can be distributed among 3 boys A, B and C such that A always gets even number of balls:

(A) 112 (B) 120

(C) 126 (D) 132

64. The area bounded by $y = x^4 + 1,$ and $y = 2x^2 - \sin^2 \pi x$ is

(A) $\frac{31}{15}$ (B) $\frac{15}{31}$

(C) $\frac{13}{31}$ (D) none of these

Space for Rough work

65. A tangent to the curve $y = 1 - x^2$ is drawn so that the abscissa x_0 of the point of tangency belongs to the interval [0, 1]. The tangent at x_0, meets the *x*-axis and *y*-axis at *A* and *B* respectively. The minimum area of the ΔOAB where *O* is the origin is

(A) $\frac{4}{9}$ (B) $\frac{4\sqrt{3}}{9}$

(C) $\frac{1}{4}$ (D) none of these

66. If $a_n = \sqrt{2+\sqrt{2+\sqrt{2}+....}}$ having n radial signs, then the value of a_n is:

(A) $2\sin\left(\frac{n}{2^{n+1}}\right)$ (B) $2\cos\left(\frac{n}{2^{n+1}}\right)$

(C) $\sqrt{2}\cos\left(2^{n+1}\pi\right)$ (D) $4\cos\left(\frac{\pi}{2^{n+1}}\right)$

67. If $x^2 - 2axy - y^2 = 0$ and $x^2 - 2bxy - y^2 = 0$ bisect the angles between each other, then the relation between 'a' and 'b' is:

(A) $a + b = 1$ (B) $ab = 1$

(C) $ab + 1 = 0$ (D) $a^2 + ab + b^2 = 0$

68. If the sides of a triangle are $z^2 + z + 1, 2z + 1$ and $z^2 - 1$, then its largest angle is:

(A) $150°$ (B) $105°$

(C) $120°$ (D) $130°$

69. If α, β be the roots of $m^2\left(x^2 - x\right) + 2mx + 3 = 0$ and m_1, m_2 be the two values of m for which α, β are connected by the relation $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{4}{3}$, then the value of $\frac{m_1^2}{m_2} + \frac{m_2^2}{m_1}$ is:

(A) $\frac{-58}{3}$ (B) $\frac{-68}{3}$

(C) $\frac{-76}{3}$ (D) $\frac{-86}{3}$

Space for Rough work

70. Let $k = \log_2 \log_3 5^{\lambda}$, then set of values of λ such that the following inequality is satisfied $1 < (25)^{2^{-k}} < 3$:

(A) $\left(\frac{-1}{2}, \infty\right)$ (B) $\left(\frac{1}{2}, \infty\right)$

(C) $(1, \infty)$ (D) $(2, \infty)$

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

71. If $\theta \varepsilon (-\pi, \pi)$, then the number of real solutions of the equation $\cos^7 \theta + \sin^4 \theta - 1 = 0$ is:

72. The number of solutions of the equation $\log_{x^2+6x+8} \log_{2x^2+2x+3} (x^2 - 2x) = 0$ is:

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

73. The number of ways in which 21 identical balls can be distributed among 3 boys A, B and C such that A always gets even number of balls:

74. If $\frac{1}{\sqrt{4x+1}}\left\{\left(\frac{1+\sqrt{4x+1}}{2}\right)^n - \left(\frac{1-\sqrt{4x+1}}{2}\right)^n\right\} = a_0 + a_1 x + ... + a_5 x^5$, then the value of n is:

75. The number of non-negative integral solutions of 3x + y + z = 24, is

Space for Rough work